

AYMANE AIT DADS

Data Science & AI Intern | AI-Driven Automation · Scalable Data Pipelines · LLM Fine-Tuning

Ayman.Ait-dads@eurecom.fr | +33 7 60 92 50 93 | linkedin.com/in/aymane-ait-dads | Sophia Antipolis, France

PROFESSIONAL SUMMARY

Achieved top 10% public leaderboard (0.80/1.0) in the \$106K+ NVIDIA Nemotron Kaggle competition by fine-tuning a 30B Mamba-Transformer MoE model with LoRA, directly demonstrating ability to build and validate non-deterministic AI automation solutions at scale. Built and deployed a clustering pipeline at Orange Maroc on 100K+ network records, reducing manual analysis time by 60% and delivering data-driven recommendations to the network optimization team. Brings hands-on experience in Python, SQL, scalable data pipelines, ETL workflows, cloud computing, and AI-enabled development with Claude API and LangChain. Available for a USA/Canada internship starting Summer 2025; visa sponsorship required (currently based in France, Moroccan national).

CORE COMPETENCIES

AI-Driven Automation | Scalable Data Pipelines | ETL Pipelines | SQL & Python | Non-Deterministic AI Validation | Cloud Computing | Machine Learning Initiatives | Data Integrity & Compliance

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built AI-driven automation pipeline combining Random Forest and K-Means clustering on **100K+ fiber-optic network records**, classifying network quality across thousands of nodes for the optimization team.
- Designed and delivered a scalable clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output consumed directly by operations stakeholders for maintenance decisions.
- Reduced manual data analysis time by **~60%** through automated feature engineering, model evaluation, and end-to-end ETL workflow integration using Python, Pandas, and Scikit-learn.
- Documented technical processes and translated machine learning model outputs into actionable maintenance recommendations via **Power BI stakeholder presentations** for cross-functional business teams.

PROJECTS

NVIDIA Nemotron Reasoning Challenge - Kaggle Competition

Kaggle · In Progress (June 2026) · Top 10% Public Leaderboard | \$106K+ Prize Pool

- Achieved **top 10% public leaderboard score of 0.80/1.0** (vs. 0.49 base, 0.86 best) fine-tuning a 30B Mamba-Transformer MoE model with LoRA ($r=32$, ~888M trainable parameters) on 7,828 chain-of-thought reasoning examples using SFTTrainer with completion-only loss.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** in strict dependency order - including monkey-patching CUDA allocators, stubbing mamba3/cutlass imports, and replacing Triton kernels with PyTorch fallbacks - and executed 20+ single-variable ablations confirming packing and synthetic data as performance degraders.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton

AI-Generated Image Detection - NTIRE 2026 @ CVPR & EURECOM ImSecu

EURECOM Course + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in EURECOM Class Leaderboard

- Ranked **1st on the EURECOM private class Kaggle leaderboard (0.791 AUC)** by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head using dual learning rates, trained on 250K images spanning 25+ generator types with FP16 mixed precision.
- Submitted independently to the **NTIRE 2026 @ CVPR CodaBench international benchmark**; validated that 1-epoch training (0.793 AUC) outperforms 4-epoch training (0.767 AUC) through ablation, and applied a 3-model ensemble with 10-view TTA to optimize out-of-distribution generalization.

PyTorch · OpenCLIP · Hugging Face Transformers · Kaggle T4 GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, Cloud Computing, NLP, Computer Vision, Web Applications, Image Security

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

LLM Fine-Tuning & AI Automation: LoRA · QLoRA · SFTTrainer · Hugging Face Transformers · Unsloth · vLLM · LangChain · Claude API · RAG · prompt engineering

Data Science & Engineering: Python · SQL · Pandas · NumPy · Scikit-learn · PyTorch · TensorFlow · feature engineering · ETL pipelines · PostgreSQL · Supabase · Docker

MLOps, Cloud & Visualization: Git · Linux · CUDA debugging · FP16/BF16 mixed precision · Kaggle GPU · Power BI · Matplotlib · Seaborn · REST APIs · ablation studies